

PROXMOX VE

Virtualization Environment

Comprehensive Study Notes for Proxmox VE Administrators

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TABLE OF CONTENTS

1. Introduction to Proxmox VE
2. Installation & Initial Configuration
3. Proxmox Web Interface (GUI)
4. Virtual Machines (KVM/QEMU)
5. Linux Containers (LXC)
6. Storage Management
7. Networking in Proxmox VE
8. Cluster & High Availability (HA)
9. Backup & Restore
10. User Management & Permissions (RBAC)
11. Firewall & Security
12. Monitoring & Performance
13. Proxmox CLI Reference
14. Troubleshooting & Best Practices

CHAPTER 1 — Introduction to Proxmox VE

What is Proxmox VE?

Proxmox Virtual Environment (Proxmox VE or PVE) is an open-source enterprise-grade server virtualization management platform. It is based on Debian Linux and integrates two virtualization technologies — KVM (Kernel-based Virtual Machine) for full virtualization and LXC (Linux Containers) for lightweight container-based virtualization. Proxmox VE provides a web-based management interface, REST API, and CLI tools for complete datacenter-level management.

Key Features of Proxmox VE

Feature	Description
KVM Virtualization	Full hardware virtualization for running any OS (Windows, Linux, BSD)
LXC Containers	Lightweight OS-level virtualization with near-native performance
Cluster Support	Up to 32 nodes in a single cluster using Corosync + pvecm
High Availability (HA)	Automatic VM/CT failover across cluster nodes
Software-Defined Storage	ZFS, Ceph, LVM, NFS, iSCSI, CIFS, GlusterFS support
Proxmox Backup Server	Incremental, space-efficient backups with deduplication
RBAC Security	Role-based access control with fine-grained permissions
Live Migration	Migrate running VMs between nodes with zero downtime
REST API	Full management automation via API and Terraform provider
Open Source	AGPLv3 license; enterprise subscription available from Proxmox GmbH

Proxmox VE vs Other Hypervisors

Parameter	Proxmox VE	VMware ESXi	Hyper-V
License	Open Source (AGPLv3)	Commercial	Commercial / Free Core
Hypervisor Type	Type 1 (Bare Metal)	Type 1 (Bare Metal)	Type 1 (Bare Metal)
Container Support	LXC built-in	No native containers	No native containers

Parameter	Proxmox VE	VMware ESXi	Hyper-V
GUI	Web-based (no client needed)	vSphere Web Client	Hyper-V Manager / WAC
HA Clustering	Built-in, free	vCenter required (paid)	Failover Cluster (complex)
Storage Backend	ZFS, Ceph, LVM, NFS++	VMFS, vSAN (paid)	CSV, Storage Spaces
Cost	Free / ~€119/year enterprise	Expensive licensing	Included in Windows Server

Proxmox VE Architecture

Debian Linux Base — Proxmox VE runs on Debian stable (currently Debian 12 Bookworm). It adds its own kernel patches, packages, and repositories on top.

Hypervisor Layer — KVM is a Linux kernel module enabling hardware-assisted virtualization (Intel VT-x / AMD-V). QEMU is the userspace emulator that works alongside KVM to simulate hardware devices.

Container Engine — LXC (Linux Containers) shares the host kernel, making containers extremely lightweight with minimal overhead compared to full VMs.

Management Stack — pvedaemon handles API requests; pveproxy serves the web GUI on port 8006 (HTTPS). Corosync handles cluster communication; pmxcfs is the cluster file system (stored in RAM, backed by corosync).

□ Important: Proxmox VE requires hardware with Intel VT-x or AMD-V (AMD-Vi) support for KVM virtualization. For LXC containers, any 64-bit CPU is sufficient.

CHAPTER 2 — Installation & Initial Configuration

System Requirements

Component	Minimum	Recommended
CPU	64-bit dual-core	64-bit quad-core or more with VT-x/AMD-V
RAM	2 GB	8 GB or more (16 GB for production)
Storage	32 GB HDD	SSD 128 GB+ (ZFS: ECC RAM recommended)
Network	1x Gigabit NIC	2x Gigabit NIC (management + VM traffic)
IPMI/iDRAC	Optional	Recommended for remote management

Installation Steps

- **1. Download ISO:** Download Proxmox VE ISO from <https://www.proxmox.com/downloads>
- **2. Create Bootable USB:** Use Rufus (Windows) or dd command (Linux) to write ISO to USB
- **3. Boot from USB:** Set USB as first boot device in BIOS/UEFI. Disable Secure Boot.
- **4. Select Disk:** Choose target HDD/SSD. Optionally select ZFS, ext4, or xfs filesystem.
- **5. Set Locale:** Choose country, timezone, and keyboard layout
- **6. Network Config:** Set management IP, gateway, DNS, and FQDN (e.g. pve01.lab.local)
- **7. Set Password:** Set root password and admin email address
- **8. Install:** Installation takes ~5–10 minutes. Remove USB and reboot.
- **9. Access GUI:** Open browser: <https://:8006> | Login: root /

Post-Installation Configuration

1. Update Package Sources (No Subscription)

By default, Proxmox points to the enterprise repository which requires a paid subscription. Edit the sources:

```
# Disable enterprise repo
echo '# deb https://enterprise.proxmox.com/debian/pve bookworm pve-enterprise' >
/etc/apt/sources.list.d/pve-enterprise.list

# Add no-subscription repo
echo 'deb http://download.proxmox.com/debian/pve bookworm pve-no-subscription' >
/etc/apt/sources.list.d/pve-no-subscription.list

apt update && apt dist-upgrade -y
```

2. Remove Subscription Nag Screen

Run the following to remove the nag popup (community use):

```
sed -i.bak 's/!res.data.active)/true)/g'  
/usr/share/javascript/proxmox-widget-toolkit/proxmoxlib.js && systemctl restart  
pveproxy
```

3. Configure NTP Time Synchronization

```
chrony -q 'server 0.pool.ntp.org iburst'  
systemctl enable --now chrony
```

Network Configuration Post-Install

Proxmox VE uses **/etc/network/interfaces** for network configuration. The default setup creates a Linux bridge {**vmbr0**} connected to the physical NIC.

```
auto lo iface lo inet loopback auto eno1 iface eno1 inet manual auto vmbr0 iface  
vmbr0 inet static address 192.168.1.100/24 gateway 192.168.1.1 bridge-ports eno1  
bridge-stp off bridge-fd 0
```

□ After editing **/etc/network/interfaces**, run: **systemctl restart networking** OR **ifreload -a**

CHAPTER 3 — Proxmox Web Interface (GUI)

Accessing the GUI

Access Proxmox VE management via browser at **https://:8006**. Use self-signed certificate (accept browser warning) or install Let's Encrypt / custom certificate. Default login: **{bold('root')}** with PAM realm.

GUI Layout Overview

Panel	Location	Purpose
Resource Tree	Left sidebar	Tree view of Datacenter ▢ Node ▢ VMs/CTs/Storage
Toolbar	Top bar	Create VM, Create CT, search, task viewer, user menu
Content Area	Center/Right	Configuration tabs for selected resource
Log Viewer	Bottom	Real-time task progress and log output
Node Summary	Node ▢ Summary	CPU, RAM, disk, network graphs; uptime

Datacenter Level Options

At the **Datacenter** level (root of resource tree), you can configure:

- Summary — Cluster overview, node status, guest list
- Search — Find VMs, containers, nodes across the cluster
- Options — Default keyboard layout, console viewer settings (noVNC / SPICE / xterm.js)
- Storage — Add and manage storage backends (local, NFS, Ceph, etc.)
- Backup — Schedule Proxmox Backup Server (PBS) or vzdump backups
- Replication — Configure ZFS replication jobs between nodes
- HA — Configure High Availability groups, resources, and fencing
- ACME — Let's Encrypt certificate management for the web GUI
- Firewall — Datacenter-wide firewall rules and security groups
- Permissions — Users, Groups, Roles, API tokens, Two-Factor Auth
- Cluster — Cluster configuration using pvecm command set

Node Level Options

Clicking on a **Node** in the tree reveals node-specific management options:

Tab	Function
Summary	Real-time CPU, memory, disk I/O, and network utilization graphs
Notes	Free-text notepad for the node (supports Markdown)
Shell	Browser-based root shell directly on the node
System	Network interfaces, DNS, hosts, time/NTP configuration
Updates	APT package management – view and install available updates
Firewall	Node-level iptables/nftables firewall rules
Disks	Physical disk list, ZFS pools, LVM groups, directory creation
Ceph	Manage Ceph OSDs, monitors, pools (if Ceph is installed)
Replication	Configure ZFS snapshot-based replication jobs
Task History	Complete log of all tasks run on the node
Subscription	View, activate, or manage Proxmox VE subscription key

CHAPTER 4 — Virtual Machines (KVM/QEMU)

Creating a Virtual Machine

Click **Create VM** in the top toolbar or right-click a node. The wizard walks through the following tabs:

Wizard Tab	Key Settings
General	Node, VM ID (100–999999), Name, Resource Pool, Start at boot
OS	ISO image source, Guest OS type (Linux/Windows/Other), version
System	Graphic card, SCSI controller (VirtIO SCSI recommended), BIOS (SeaBIOS/OVMF), Qemu Agent
Disks	Bus/Device (virtio0), Storage pool, Size, Cache mode, Discard (SSD trim)
CPU	Sockets, Cores, Type (host=best perf), NUMA, CPU limit
Memory	RAM size in MiB, Ballooning device (dynamic memory), Min memory
Network	Bridge (vmbro0), Model (VirtIO=best), VLAN tag, MAC address, Firewall
Confirm	Review settings; optionally check 'Start after created'

VM Configuration Settings (Important)

CPU Types

CPU Type	Use Case
host	Best performance — exposes all host CPU flags. Cannot live-migrate between different CPU generations.
kvm64	Default — maximum compatibility, safe for cross-node migration
Haswell/Broadwell/SkyLake	Intel-specific CPU models for specific instruction set requirements
max	Like 'host' but with additional virtual features; not for production clusters

Disk Cache Modes

Cache Mode	Description	Best For
none	No cache — I/O directly to storage; uses O_DIRECT	Production (recommended)

Cache Mode	Description	Best For
writeback	Write cache enabled — faster writes, risk of data loss on power failure	Performance testing
writethrough	Write goes to both cache and storage — safe but slower	Safe environments
directsync	Same as writethrough without read cache	Max data integrity
unsafe	All caching enabled — fastest, but dangerous	Lab/testing only

QEMU Guest Agent

The **QEMU Guest Agent** is a service running inside the VM that allows Proxmox to communicate with the guest OS for operations like graceful shutdown, frozen filesystem snapshots, and IP address retrieval.

Install inside Linux VM:

```
apt install qemu-guest-agent && systemctl enable --now qemu-guest-agent
```

Install inside Windows VM: Download and install VirtIO Windows drivers ISO (stable-virtio ISO from Fedora project).

VM Management Commands

Task	Command
Start VM	qm start
Stop VM (graceful)	qm shutdown
Stop VM (force)	qm stop
Reset VM	qm reset
VM Status	qm status
List all VMs	qm list
VM Config	qm config
Clone VM	qm clone --name --full
Create Template	qm template
Delete VM	qm destroy
Migrate VM	qm migrate

Task	Command
Resize Disk	qm resize +G
Snapshot	qm snapshot
Rollback Snapshot	qm rollback

CHAPTER 5 — Linux Containers (LXC)

Containers vs VMs in Proxmox

Aspect	LXC Container	KVM Virtual Machine
Kernel	Shares host kernel	Has own kernel (full isolation)
Overhead	~50–100 MB RAM per CT	~512 MB+ RAM per VM overhead
Boot Time	1–3 seconds	30–60 seconds
OS Support	Linux only	Any OS (Windows, BSD, Linux)
Isolation	OS-level isolation	Full hardware isolation
Best For	Linux services, microservices, web servers	Windows guests, legacy apps, full OS environments

Creating an LXC Container

Click **Create CT** from the toolbar. Container templates must first be downloaded from the Proxmox template repository via Storage ▢ CT Templates ▢ Templates.

Tab	Key Configuration
General	Node, CT ID, Hostname, Password (root), SSH public key, Unprivileged CT toggle
Template	Select downloaded OS template (Ubuntu, Debian, Alpine, CentOS, etc.)
Disks	Storage pool and rootfs size
CPU	Number of CPU cores
Memory	RAM and Swap in MiB
Network	Interface name, bridge, IP config (DHCP or static), VLAN
DNS	Search domain and DNS servers

Privileged vs Unprivileged Containers

Unprivileged Containers (recommended): UID/GID remapping — container root (UID 0) maps to an unprivileged host user (e.g., UID 100000). Even if a container is compromised, the attacker only has unprivileged host access. Use unprivileged for all production workloads.

Privileged Containers: Container root maps to host root (UID 0). Required for some advanced use cases (NFS, Docker inside LXC, USB passthrough). Use only when necessary and with proper security controls.

LXC Container Commands

Task	Command
Start CT	pct start
Stop CT	pct stop
Shutdown CT	pct shutdown
Enter CT Shell	pct enter
List all CTs	pct list
CT Config	pct config
Clone CT	pct clone --hostname
Delete CT	pct destroy
Resize Disk	pct resize rootfs +5G
Snapshot CT	pct snapshot
Rollback	pct rollback
Push file to CT	pct push
Pull file from CT	pct pull

CHAPTER 6 — Storage Management

Storage Types in Proxmox VE

Storage Type	Protocol	VM Disks	ISO/Templates	Shared
Directory	Local filesystem	□ (raw/qcow2/vmdk)	□	□
LVM	Block device	□ (raw)	□	□
LVM-Thin	Block (thin pools)	□ (raw/snapshot)	□	□
ZFS (local)	Local ZFS pool	□ (zvol/subvol)	□	□
NFS	Network (NFS)	□ (qcow2/vmdk)	□	□
CIFS/SMB	Network (SMB)	□ (qcow2/vmdk)	□	□
iSCSI	Block (network)	□ (raw via LUN)	□	□
Ceph/RBD	Distributed block	□ (raw/RBD)	□	□
CephFS	Distributed FS	□ (qcow2)	□	□
PBS (Proxmox Backup Server)	Custom	Backups only	□	□

ZFS Storage (Recommended for Production)

ZFS is a copy-on-write filesystem with built-in RAID, compression, deduplication, and snapshot capabilities. Proxmox VE has first-class ZFS support. **ECC RAM is strongly recommended** when using ZFS to prevent data corruption.

Common ZFS RAID Levels:

ZFS Level	Description	Min Disks	Fault Tolerance
stripe (RAID-0)	Striped, no redundancy — max performance	2	0 disks
mirror (RAID-1)	Mirrored — data on all disks	2	N-1 disks
raidz1 (RAID-5)	Single parity — 1 disk tolerance	3	1 disk
raidz2 (RAID-6)	Double parity — 2 disk tolerance	4	2 disks
raidz3	Triple parity — 3 disk tolerance	5	3 disks

Useful ZFS Commands:

```
zpool status # View all ZFS pool status
zpool list # List pools with usage
zfs list # List all ZFS datasets
zfs snapshot pool/data@snap1 # Create snapshot
zfs rollback pool/data@snap1 # Rollback to snapshot
zpool scrub # Start integrity check
zfs set compression=lz4 # Enable LZ4 compression
```

Disk Image Formats

Format	Description	Snapshots	Best Used With
raw	Plain binary image, no overhead	LVM/ZFS-level only	LVM, ZFS, Ceph
qcow2	QEMU native format, thin provisioned	Built-in native	Directory, NFS storage
vmdk	VMware format	Limited	Compatibility/migration

CHAPTER 7 — Networking in Proxmox VE

Network Components

Component	Description
Physical NIC (enpXsY/ethX)	Physical network interface — do not assign IPs directly; use bridges
Linux Bridge (vbr0)	Software switch connecting VMs/CTs to network. Bridge can have a physical NIC as an uplink.
VLAN Interface	Tagged sub-interface (e.g., vbr0.100) for VLAN 100
Bond (LACP)	NIC teaming for redundancy and/or throughput. Modes: active-backup, LACP (802.3ad), balance-rr
OVS Bridge	Open vSwitch-based bridge — advanced SDN features (requires openvswitch-switch package)
SDN (Software Defined Networking)	Proxmox built-in SDN with VXLAN, VRF, EVPN for multi-node virtual networks

VLAN Configuration Example

Proxmox bridges support VLAN-aware mode (single bridge, multiple VLANs) or traditional separate bridge-per-VLAN. Enable VLAN-aware on the bridge:

```
auto vbr0 iface vbr0 inet static address 192.168.1.10/24 gateway 192.168.1.1
bridge-ports eno1 bridge-stp off bridge-fd 0 bridge-vlan-aware yes bridge-vids
2-4094
```

Assign VLAN tag to VM in VM ▢ Network ▢ VLAN Tag field.

NIC Bonding (Link Aggregation)

Bond two physical NICs for redundancy (**active-backup**) or throughput (**802.3ad LACP**):

```
auto bond0 iface bond0 inet manual bond-slaves eno1 eno2 bond-miimon 100
bond-mode active-backup auto vbr0 iface vbr0 inet static address
192.168.1.10/24 gateway 192.168.1.1 bridge-ports bond0 bridge-stp off bridge-fd
0
```

Software Defined Networking (SDN)

Proxmox SDN (available from PVE 7.x) allows creating isolated virtual networks spanning multiple cluster nodes using VXLAN overlays. Configure via **Datacenter ▢ SDN**.

SDN Object	Purpose
Zone	Defines the network type (Simple, VLAN, QinQ, VXLAN, EVPN)

SDN Object	Purpose
VNet	A virtual network (like a VLAN) within a zone
Subnet	IP subnet with optional DHCP and gateway for a VNet

CHAPTER 8 — Cluster & High Availability (HA)

Proxmox Cluster Basics

A Proxmox cluster allows multiple nodes to be managed from a single interface and enables VM/CT migration and HA. Cluster communication uses **Corosync** (UDP port 5405) and the cluster filesystem **pmxcfs** (stored in RAM, backed to /etc/pve).

Requirement	Detail
Minimum nodes	3 nodes recommended for quorum; 2 nodes requires external quorum device (QDevice)
Network	Low-latency dedicated network for cluster communication strongly recommended
Same Proxmox VE	All nodes should run the same PVE major version
Shared Storage	Required for live migration; local storage limits migration options
Time Sync	All nodes must have synchronized time (NTP/chrony)

Cluster Setup Commands

On the first node (cluster creator):

```
pvecm create
```

On each additional node (joining):

```
pvecm add
```

Verify cluster status:

```
pvecm status
```

```
pvecm nodes
```

```
corosync-cfgtool -s
```

High Availability (HA)

Proxmox HA monitors VMs and CTs. If a node fails, the HA manager automatically restarts affected guests on surviving nodes. HA requires **at least 3 nodes** or a QDevice for quorum.

HA Components:

Component	Role
HA Manager	pve-ha-lrm (Local Resource Manager) + pve-ha-crm (Cluster Resource Manager)

Component	Role
Watchdog	Hardware or software watchdog — ensures fencing works. Default: /dev/watchdog
Fencing	Kills an unresponsive node to prevent split-brain (requires IPMI/iDRAC or watchdog)
HA Groups	Assign VMs to specific nodes with priority settings
HA Resources	VMs/CTs registered in HA with states: started, stopped, migrated, error

HA States:

- **started:** Resource should be running — HA will start it if stopped
- **stopped:** Resource should be stopped — HA will stop it if running
- **disabled:** HA ignores this resource
- **ignored:** Like disabled but with no fence action

Live Migration

Live migration moves a running VM between nodes with minimal downtime.

Requirements: **shared storage** (NFS, Ceph, iSCSI) for the VM disk, same CPU architecture, and cluster membership.

```
qm migrate --online 1
```

For local storage (non-shared), use offline migration or --with-local-disks flag (copies disk):

```
qm migrate --online 1 --with-local-disks
```

CHAPTER 9 — Backup & Restore

Backup Methods

Method	Description	Best For
vzdump (built-in)	Proxmox native backup tool; supports stop/suspend/snapshot modes	Quick local/NFS backups
Proxmox Backup Server (PBS)	Dedicated backup server with deduplication, incremental backups, encryption	Production enterprise environments
ZFS Snapshots	Instant, space-efficient disk-level snapshots within ZFS pools	ZFS storage users
VM/CT Snapshots	In-memory or disk-only snapshots via Proxmox GUI (qcow2/ZFS)	Quick state capture

vzdump Backup Modes

Mode	Description	Downtime	Consistency
stop	Stops VM, backs up, restarts	Yes (brief)	100% consistent
suspend	Suspends VM, backs up, resumes	Yes (brief)	Consistent
snapshot	Live backup using disk snapshot	None	Crash-consistent (recommended)

vzdump Command Examples

```
# Backup VM 100 to local storage in snapshot mode
vzdump 100 --storage local --mode snapshot --compress zstd

# Backup all VMs on node
vzdump --all --storage nfs-backup --mode snapshot --compress zstd

# Backup VM to Proxmox Backup Server
vzdump 100 --storage pbs-storage --mode snapshot
```

Restore from Backup

Restore via GUI: Storage ▢ Backups ▢ Select .vma file ▢ Restore.

Restore via CLI:

```
# Restore VM from vzdump archive qmrestore
/var/lib/vz/dump/vzdump-qemu-100-*.vma.zst 101 --storage local-lvm
```

```
# Restore CT pct restore 102 /var/lib/vz/dump/vzdump-lxc-102-*.tar.zst --storage local-lvm
```

Scheduled Backups

Configure automated backup schedules at **Datacenter ▢ Backup**. Specify node, storage, days/time, guest selection, retention policy (keep last N backups), and email notification.

▢ Backup Retention Options: Keep Last N copies, Keep Last N daily/weekly/monthly/yearly – configure via Datacenter ▢ Backup ▢ Edit ▢ Retention tab.

CHAPTER 10 — User Management & RBAC

Realms (Authentication Sources)

Realm	Description
PAM (Linux PAM)	Local Linux system users — only for root and local OS accounts
PVE (Built-in)	Proxmox-native user database — recommended for non-root admin users
LDAP	Authenticate users from an LDAP directory (OpenLDAP)
Active Directory	Authenticate against Microsoft Active Directory (AD DS)
OpenID Connect	OAuth2/OIDC integration (Google, Keycloak, etc.) — PVE 7+

Built-in Roles

Role	Permissions Level
Administrator	Full access — equivalent to root permissions on all resources
PVEAdmin	Most admin rights except system-level config
PVEVMAdmin	Manage VMs — create, configure, start/stop, console; no node/storage admin
PVEVMUser	Use VMs — console, start/stop; no configuration changes
PVEAuditor	Read-only access to all resources
PVEDatastoreAdmin	Manage storage, backups, templates
PVEDatastoreUser	Allocate space, backup/restore own VMs
PVENetworkAdmin	Manage network configuration
NoAccess	Explicitly deny all access

Permission Path Examples

Permissions in Proxmox follow a hierarchical path structure. A permission at a higher path propagates down unless overridden.

Permission Path	Scope
/	Root — all resources (Datacenter level)
/nodes/	Specific node only
/vms/	Specific VM only

Permission Path	Scope
/storage/	Specific storage backend
/pool/	All resources in a resource pool

API Tokens

API Tokens allow automation without using root credentials. Tokens belong to a user and can have the **Privilege Separation** flag — limiting token permissions to a subset of the owning user's privileges. Create via: **Datacenter** ▢ **Permissions** ▢ **API Tokens** ▢ **Add**.

API authentication header:

```
Authorization: PVEAPIToken=user@realm!tokenid=
```

Two-Factor Authentication (2FA)

Proxmox supports TOTP (Time-based One-Time Password) and WebAuthn (hardware security keys) for user accounts. Enable via Datacenter ▢ Permissions ▢ Two-Factor Auth ▢ Add TOTP.

CHAPTER 11 — Firewall & Security

Proxmox Firewall Levels

Proxmox VE has a 3-tier firewall system using nftables (PVE 8+) / iptables (PVE 7).

Level	Config Location	Applies To
Datacenter	/etc/pve/firewall/cluster.fw	Global rules, IPSets, Security Groups
Node	/etc/pve/nodes//host.fw	Node host system traffic
VM/CT	/etc/pve/firewall/.fw	Traffic to/from a specific guest

Firewall Rule Parameters

Parameter	Values / Example
Direction	IN (ingress) or OUT (egress)
Action	ACCEPT, DROP, REJECT
Protocol	tcp, udp, icmp, any
Source	IP, CIDR, IPSet name, Security Group name
Destination	IP, CIDR, IPSet name, Security Group name
Dest Port	Port number or range (80, 443, 8006, 22:25)
Comment	Human-readable description
Enable	Toggle rule on/off without deleting
Macro	Pre-defined rules: SSH, HTTP, HTTPS, Ping, etc.

Security Hardening Best Practices

- Change default SSH port or restrict SSH access by IP via firewall rules
- Disable root login over SSH: PermitRootLogin no in /etc/ssh/sshd_config (create a sudo admin instead)
- Enable Two-Factor Authentication (TOTP) for all GUI admin accounts
- Use a separate management VLAN for Proxmox node IP — isolate from VM traffic
- Keep Proxmox VE updated: apt update && apt dist-upgrade regularly
- Use fail2ban to prevent brute-force attacks on SSH and the GUI (port 8006)
- Restrict web GUI access: firewall rule to allow port 8006 only from admin IPs
- Use read-only API tokens for monitoring tools — never give monitoring full admin access

- Enable firewall at datacenter, node, and VM levels with default DROP policy
- Regularly audit user accounts and remove unused ones
- Configure automated backups and test restores periodically

CHAPTER 12 — Monitoring & Performance

Built-in Monitoring

Proxmox VE includes RRDtool-based graphs for CPU, memory, disk I/O, and network for each node, VM, and container. Available under: Node/VM/CT ▢ Summary.

External Monitoring Integration

Tool	Integration Method
Prometheus + Grafana	Use pve-exporter (Python) or pveum API to scrape PVE metrics
InfluxDB	Configure via Datacenter ▢ Options ▢ Metric Server ▢ InfluxDB
Graphite	Configure via Datacenter ▢ Options ▢ Metric Server ▢ Graphite
Zabbix	Proxmox Zabbix template available for host and guest monitoring
Netdata	Install on each Proxmox node for real-time monitoring

Key Performance Commands

Command	Purpose
top / htop	Real-time CPU and process monitoring
iostat -xz 1	Disk I/O statistics per device
iotop	Per-process disk I/O
pvesh get /nodes//status	Node resource usage via API
qm monitor	QEMU monitor console for VM performance info
zpool iostat 1	ZFS pool I/O statistics
cat /proc/meminfo	Memory usage details
dmesg grep -i error	Kernel error messages
journalctl -u pveproxy	Proxmox web proxy logs
journalctl -u pvedaemon	Proxmox daemon logs

InfluxDB Metrics Configuration

Navigate to **Datacenter ▢ Options ▢ Metric Server**. Click Add ▢ InfluxDB. Specify server IP, port (8086), organization, bucket, and API token. Proxmox will push host and guest metrics automatically every minute.

CHAPTER 13 — Proxmox CLI Reference

Essential CLI Commands

Command	Description
pvesh	CLI wrapper for Proxmox REST API (explore: pvesh ls /)
pvecm	Cluster management (create, add, status, nodes)
pveum	User management (user, group, role, acl, token)
pvesm	Storage management (add, remove, list, status)
pveam	Appliance/template management (update, download, list)
pvenode	Node management (wakeonlan, config, startall, stopall)
pve-ssl-renew	Renew ACME/Let's Encrypt certificates
ha-manager	HA manager control (status, enable, disable, migrate)
pvesubscription	Manage PVE subscription key

pvesh Examples

```
pvesh get /nodes # List all nodes
pvesh get /nodes/pve01/status # Node resource usage
pvesh get /nodes/pve01/qemu # List VMs on node
pvesh get /cluster/resources # All cluster resources
pvesh create /nodes/pve01/qemu/100/status/start # Start VM 100
```

pveum Examples (User Management CLI)

```
pveum user list # List all users
pveum user add admin@pve --password Pass123 # Create user
pveum user modify admin@pve --groups admins # Add to group
pveum acl modify / --users admin@pve --roles Administrator # Grant role
pveum token add admin@pve api-token # Create API token
pveum role list # List all roles
```

pvesm Examples (Storage CLI)

```
pvesm status # Storage status overview
pvesm list local # List content in 'local' storage
pvesm add nfs nfs-backup --server 192.168.1.5 --export /backups --content backup,iso
pvesm remove nfs-backup # Remove storage
```

System Service Commands

Service	Command
Web GUI	systemctl restart pveproxy
PVE Daemon	systemctl restart pvedaemon
Cluster FS	systemctl restart pve-cluster
Corosync	systemctl restart corosync
HA LRM	systemctl restart pve-ha-lrm
HA CRM	systemctl restart pve-ha-crm
Firewall	systemctl restart pve-firewall
QEMU/KVM	systemctl status qemu-server
LXC	systemctl status lxc

CHAPTER 14 — Troubleshooting & Best Practices

Common Issues & Solutions

Problem	Cause	Solution
Cannot access GUI after reboot	pveproxy service failed or firewall block	SSH in, run: systemctl restart pveproxy; check port 8006 firewall
VM stuck in 'locked' state	Failed task or crash left lock	qm unlock
Cluster shows quorum lost	Node offline, corosync ring failure	Check corosync: corosync-cfgtool -s; verify network between nodes
High storage I/O / slow VMs	Disk cache mode, no SSD TRIM, network storage issue	Use cache=none for disk; enable discard; check storage latency
VM has no network after live migration	Bridge name differs on target node	Ensure all nodes have identical bridge names (vmbr0)
LXC container fails to start	Storage not mounted, template corrupt, disk full	pct start 2>&1; check journalctl; verify storage available
ZFS pool degraded	Disk failure	zpool status; replace faulty disk: zpool replace
Cannot create snapshot	VM has passthrough devices or memory ballooning issues	Disable ballooning or passthrough; use qcow2 disk format

Best Practices Summary

Infrastructure

- Use at least 3 nodes for production clusters to ensure quorum
- Separate management, VM/CT, storage, and corosync on different NICs or VLANs
- Use 10GbE networking for storage traffic (Ceph, iSCSI, NFS)
- Use ECC RAM for all ZFS nodes to prevent data corruption

Virtual Machines

- Always install QEMU Guest Agent inside VMs for better integration
- Use VirtIO drivers for network and disk in Linux/Windows for best performance
- Use CPU type 'x86-64-v2-AES' or 'host' for production; 'kvm64' for migration flexibility
- Enable Start at Boot for critical VMs; set HA for automatic failover

Storage

- Use ZFS with RAID-Z2 for local storage in production (2-disk fault tolerance)
- Enable LZ4 compression on ZFS: `zfs set compression=lz4`
- Run ZFS scrubs monthly: `zpool scrub`
- Store ISOs and backups on separate storage from VM disks

Security

- Follow least-privilege: create dedicated users with minimal required roles
- Enable 2FA for all admin accounts
- Restrict management GUI (port 8006) access to admin IPs only via firewall
- Regularly update Proxmox VE and guest OS packages

Backup

- Test restores quarterly — a backup you haven't tested is not a backup
- Follow 3-2-1 rule: 3 copies, 2 different media, 1 offsite
- Use Proxmox Backup Server (PBS) for deduplication and encryption
- Set backup retention: at minimum keep 7 daily, 4 weekly, 12 monthly

*These study notes were compiled by **Akhilesh Chaudhari** | TechShiksha | Senior IT Engineer | Proxmox VE Administration
Reference 2025*